

1.

$$\begin{aligned}
 \left(1 + \frac{1}{n} \Omega\right)^n &= \sum_{p=0}^n \frac{n!}{p!(n-p)!} \left(\frac{1}{n} \Omega\right)^p \\
 &= \sum_{p=0}^n \frac{1}{p!} n(n-1) \dots (n-p+1) \frac{1}{n^p} \Omega^p \\
 &= \sum_{p=0}^n \frac{1}{p!} \left(1 - \frac{1}{n}\right) \left(1 - \frac{2}{n}\right) \dots \left(1 - \frac{p-1}{n}\right) \Omega^p
 \end{aligned}$$

$$\lim_{n \rightarrow \infty} \left(1 + \frac{1}{n} \Omega\right)^n = \sum_{p=0}^{\infty} \frac{1}{p!} \Omega^p = e^{\Omega}$$

Alternatively:

$$\begin{aligned}
 e^{\Omega} &= \left(e^{\frac{1}{n} \Omega}\right)^n \\
 &= \left[1 + \frac{1}{n} \Omega + \frac{1}{2} \left(\frac{1}{n} \Omega\right)^2 + \dots\right]^n \\
 &= \left(1 + \frac{1}{n} \Omega\right)^n + o\left(\frac{1}{n}\right). \text{ Therefore,} \\
 e^{\Omega} &= \lim_{n \rightarrow \infty} \left(1 + \frac{1}{n} \Omega\right)^n.
 \end{aligned}$$

2. Let $V_1, V_2 \in \mathbb{V}$, and consider

$$V = \alpha_1 V_1 + \alpha_2 V_2 \in \mathbb{V}$$

$$\begin{aligned}\langle V' | V' \rangle &= \langle V | U^\dagger U | V \rangle \\&= |\alpha_1|^2 \langle V_1 | U^\dagger U | V_1 \rangle + |\alpha_2|^2 \langle V_2 | U^\dagger U | V_2 \rangle \\&\quad + \alpha_1^* \alpha_2 \langle V_1 | U^\dagger U | V_2 \rangle + \alpha_1 \alpha_2^* \langle V_2 | U^\dagger U | V_1 \rangle \\&= \langle V | V \rangle = |\alpha_1|^2 \langle V_1 | V_1 \rangle + \\&\quad + |\alpha_2|^2 \langle V_2 | V_2 \rangle + \alpha_1^* \alpha_2 \langle V_1 | V_2 \rangle \\&\quad + \alpha_1 \alpha_2^* \langle V_2 | V_1 \rangle.\end{aligned}$$

$$\text{Since } \langle V_1 | V_1 \rangle = \langle V_1 | U^\dagger U | V_1 \rangle$$

$$\text{and } \langle V_2 | V_2 \rangle = \langle V_2 | U^\dagger U | V_2 \rangle,$$

we have

$$\begin{aligned}&\alpha_1^* \alpha_2 \langle V_1 | U^\dagger U | V_2 \rangle + \alpha_1 \alpha_2^* \langle V_2 | U^\dagger U | V_1 \rangle \\&= \alpha_1^* \alpha_2 \langle V_1 | V_2 \rangle + \alpha_1 \alpha_2^* \langle V_2 | V_1 \rangle\end{aligned}$$

for any $V_1, V_2 \in \mathbb{V}$, and any α_1, α_2 .

Therefore

$$\langle v_1, U^+ U v_2 \rangle = \langle v_1, v_2 \rangle$$

for any $v_1, v_2 \in V$.

Therefore

$$U^+ U v_2 = v_2$$

for any $v_2 \in V$.

Therefore

$$U^+ U = I$$

13. 12

$$[x, p] = \hbar i$$

$$[x, p^n] = \sum_{j=1}^n p^{j-1} [x, p] p^{n-j}$$

$$= n \hbar i p^{n-1}$$

Let $F(p) = \sum_m F_m p^m$. Then

$$[x, F(p)] = \sum_m F_m [x, p^m]$$

$$= \hbar i \sum_m F_m m p^{m-1} = \hbar i \frac{dF}{dp}$$

b. Consider

$$[x, e^{\frac{i}{\hbar} x_0 p}] = \hbar i \frac{i}{\hbar} x_0 e^{\frac{i}{\hbar} x_0 p}$$

$$= -x_0 e^{\frac{i}{\hbar} x_0 p}$$

Let

$$X|x\rangle = x|x\rangle$$

Then

$$\begin{aligned} X e^{\frac{i}{\hbar} x_0 P} |x\rangle &= e^{\frac{i}{\hbar} x_0 P} (X - x_0) |x\rangle \\ &= (x - x_0) e^{\frac{i}{\hbar} x_0 P} |x\rangle \end{aligned}$$

Therefore

$$e^{\frac{i}{\hbar} x_0 P} |x\rangle = |x - x_0\rangle$$

Consider

$$\psi(x) = \langle x | \psi \rangle$$

$$\begin{aligned} \langle x | e^{-\frac{i}{\hbar} x_0 P} | \psi \rangle &= \langle x - x_0 | \psi \rangle \\ &= \psi(x - x_0) \end{aligned}$$

$$\begin{aligned} \sum_{n=0}^{\infty} \frac{1}{n!} \left(-\frac{i}{\hbar} x_0 \right)^n \langle x | P^n | \psi \rangle \\ = \sum_{n=0}^{\infty} \frac{1}{n!} (-x_0)^n \frac{d^n \psi}{dx^n} \end{aligned}$$

Identifying terms of order x_0^n
on both sides, we have:

$$\langle x | P^n | \psi \rangle = \left(\frac{\hbar}{i} \frac{d}{dx} \right)^n \psi(x) .$$

c.

$$\frac{\hbar}{i} \frac{d}{dx} \langle x | p \rangle = \langle x | P | p \rangle$$

$$= p \langle x | p \rangle$$

Therefore

$$\langle x | p \rangle = N e^{\frac{i}{\hbar} p x}$$

S4.2.1.

$$L_x = \frac{1}{\sqrt{2}} \begin{pmatrix} 0 & 1 & 0 \\ 1 & 0 & 1 \\ 0 & 1 & 0 \end{pmatrix}, \quad L_y = \frac{1}{\sqrt{2}} \begin{pmatrix} 0 & -i & 0 \\ i & 0 & -i \\ 0 & i & 0 \end{pmatrix}, \quad L_z = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & -1 \end{pmatrix}$$

a. 1, 0 and -1

b. $|L_z=1\rangle = \begin{pmatrix} 1 \\ 0 \\ 0 \end{pmatrix}$. In that state

$$\langle L_x \rangle = (100) \frac{1}{\sqrt{2}} \begin{pmatrix} 0 & 1 & 0 \\ 1 & 0 & 1 \\ 0 & 1 & 0 \end{pmatrix} \begin{pmatrix} 1 \\ 0 \\ 0 \end{pmatrix} = 0$$

$$\langle L_x^2 \rangle = (100) \frac{1}{2} \begin{pmatrix} 1 & 0 & 1 \\ 0 & 2 & 0 \\ 1 & 0 & 1 \end{pmatrix} \begin{pmatrix} 1 \\ 0 \\ 0 \end{pmatrix} = \frac{1}{2}$$

$$\Delta L_x = \sqrt{\langle L_x^2 \rangle - \langle L_x \rangle^2} = \frac{1}{\sqrt{2}}$$

c. $|L_z=-1\rangle = \begin{pmatrix} 0 \\ 0 \\ 1 \end{pmatrix}$.

The eigenvalues of L_x are given by

$$\det(L_x - \lambda_x \mathbb{I}) = \det \begin{pmatrix} -\lambda_x & \frac{1}{\sqrt{2}} & 0 \\ \frac{1}{\sqrt{2}} & -\lambda_x & \frac{1}{\sqrt{2}} \\ 0 & \frac{1}{\sqrt{2}} & -\lambda_x \end{pmatrix} = 0$$

$$-\lambda_x \left(\lambda_x^2 - \frac{1}{2} \right) - \frac{1}{\sqrt{2}} \left(\frac{1}{\sqrt{2}} \right) (-\lambda_x)$$

$$= -\lambda_x (\lambda_x^2 - 1) = 0$$

$$\lambda_x = 0 \quad |L_x = 0\rangle = \frac{1}{\sqrt{2}} \begin{pmatrix} -1 \\ 0 \\ +1 \end{pmatrix}$$

$$\lambda_x = 1 \quad |L_x = 1\rangle = \frac{1}{2} \begin{pmatrix} 1 \\ \sqrt{2} \\ 1 \end{pmatrix}$$

$$\lambda_x = -1 \quad |L_x = -1\rangle = \frac{1}{2} \begin{pmatrix} 1 \\ -\sqrt{2} \\ 1 \end{pmatrix}$$

d. In the state $|L_z = -1\rangle = \begin{pmatrix} 0 \\ 0 \\ 1 \end{pmatrix}$, a

measurement of L_x has outcome

0 with probability $\frac{1}{2}$

1 with probability $\frac{1}{4}$

-1 with probability $\frac{1}{4}$

i.e.

$$|\psi\rangle = \frac{1}{2} \begin{pmatrix} 1 \\ 1 \\ \sqrt{2} \end{pmatrix}, \quad L_z^2 = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 1 \end{pmatrix}$$

After measuring $L_z^2 = 1$, the state is

$$\begin{aligned} N \sum_{\alpha=\pm} |L_z = 1, \alpha\rangle \langle L_z = 1, \alpha | \psi \rangle \\ = N \left(|L_z = 1\rangle \langle L_z = 1 | \psi \rangle + |L_z = -1\rangle \langle L_z = -1 | \psi \rangle \right) \\ = N \left(\frac{1}{2} \begin{pmatrix} 1 \\ 0 \\ 0 \end{pmatrix} + \frac{1}{\sqrt{2}} \begin{pmatrix} 0 \\ 0 \\ 1 \end{pmatrix} \right) = N \begin{pmatrix} \frac{1}{2} \\ 0 \\ \frac{1}{\sqrt{2}} \end{pmatrix} \end{aligned}$$

$$N^2 \left(\frac{1}{4} + \frac{1}{2} \right) = \frac{3}{4} N^2 = 1 \quad N = \frac{2}{\sqrt{3}}$$

After measuring $L_z^2 = 1$, the state is

$$\begin{pmatrix} \frac{1}{\sqrt{3}} \\ 0 \\ \frac{2}{\sqrt{3}} \end{pmatrix}$$

The probability of obtaining $L_z^2 = 1$

was

$$\frac{1}{4} + \frac{1}{2} = \frac{3}{4}$$

in state $|\psi\rangle$
Measuring L_z has outcome

1 with probability $\frac{1}{4}$

0 " " $\frac{1}{4}$

-1 " " $\frac{1}{2}$

P. The states $\{|L_z=1\rangle, |L_z=0\rangle, |L_z=-1\rangle\}$
form a complete orthonormal set. Thus

$$|\psi\rangle = c_1 |L_z=1\rangle + c_0 |L_z=0\rangle + c_{-1} |L_z=-1\rangle$$

$$P(L_z=1) = |\langle L_z=1 | \psi \rangle|^2 = |c_1|^2 = \frac{1}{4}$$

$$\text{Similarly } |c_0|^2 = \frac{1}{2} \text{ and } |c_{-1}|^2 = \frac{1}{2}$$

Thus

$$|\psi\rangle = \frac{1}{2} e^{i\delta_1} |L_z=1\rangle + \frac{1}{\sqrt{2}} e^{i\delta_0} |L_z=0\rangle \\ + \frac{1}{\sqrt{2}} e^{i\delta_{-1}} |L_z=-1\rangle$$